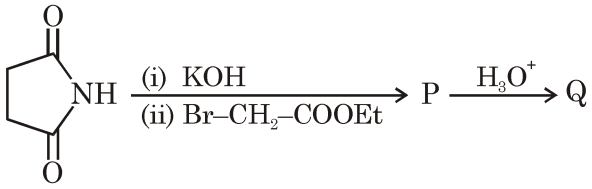
SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. Amino acid (Q) obtain in following sequence is :

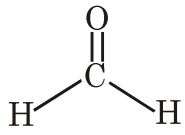


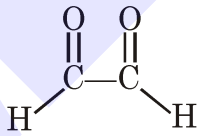
(A) Alanine (B) Valine

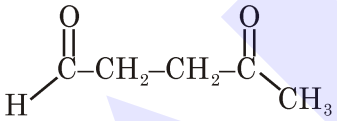
(C) Glycine (D) Leucine

2. 1,4-addition polymer of $CH_2=\underset{\displaystyle CH_3}{\overset{|}{C}}-CH=CH_2$ monomer was subjected to reductive ozonolysis.

The product formed is :

(A) 

(B) 

(C) 

(D) All correct

3. In Aniline & water mixture, Aniline can be seperate by :

(A) Steam distillation

(B) Fractional distillation

(C) Simple distillation

(D) Distillation under reduced pressure

4. The best reaction sequence to convert 2-methyl-1-bromopropane into 4-methyl-2 bromopentane is :

(A) (i) Mg in ether (ii) acetaldehyde (iii) NH_4Cl ; (iv) $SOBr_2$

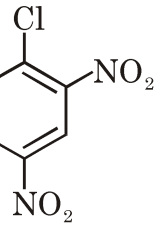
(B) (i) $NaC\equiv CH$ in ether (ii) $H_3O^+ + HgSO_4$ (iii) HBr, heat

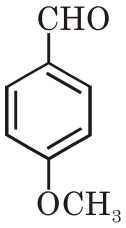
(C) (i) Alcoholic KOH (ii) CH_3COOOH (iii) H_2 / Pt (iv) HBr, heat

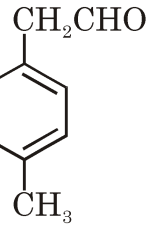
(D) (i) $NaC\equiv CH$ in ether (ii) H_2, Lindlar catalyst (iii) HCl , peroxide

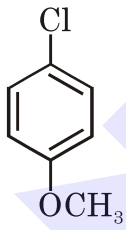
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5. The compound(s) that will **NOT** react(s) with hot concentrated aqueous alkali at atmospheric pressure is/are :

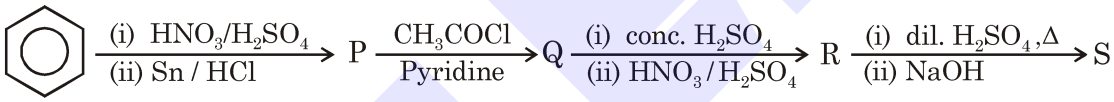
(A) 

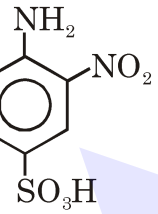
(B) 

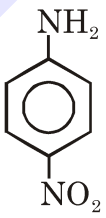
(C) 

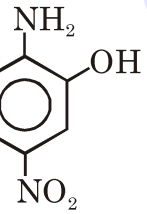
(D) 

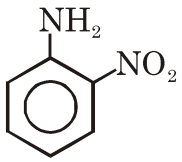
6. Major product (S) of following reaction is :



(A) 

(B) 

(C) 

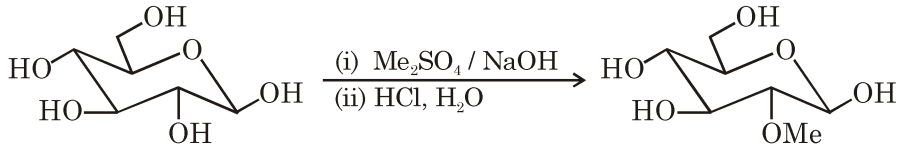
(D) 

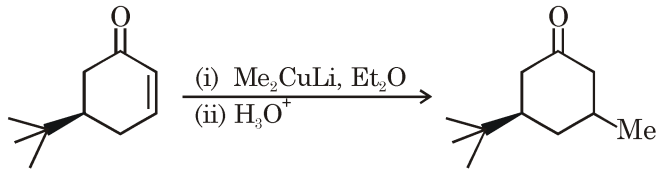
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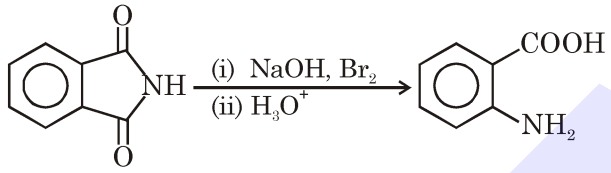
SECTION-I(ii) : One or More than one

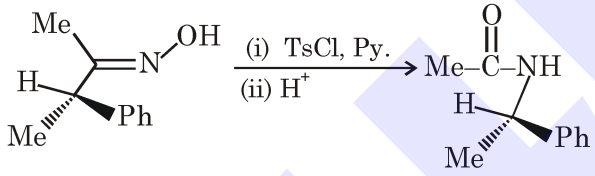
This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

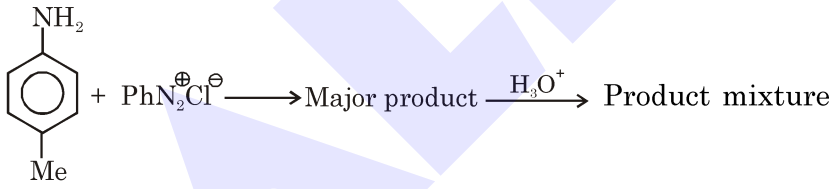
7. Identify reaction correctly matched with their major product ?

(A) 

(B) 

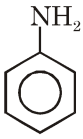
(C) 

(D) 

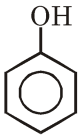
8. 

Which of the following is/are present in final reaction mixture :

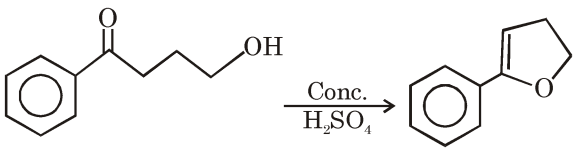
(A) 

(B) 

(C) 

(D) 

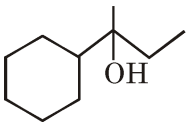
9. Which of the following reaction(s) / intermediate are involved in following reaction :



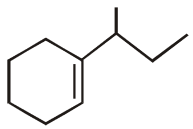
(A) Nucleophilic substitution
(B) Elimination
(C) Nucleophilic addition
(D) Dehydration

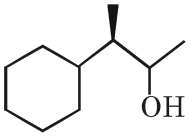
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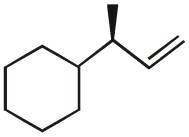
10. Which of the following option can produce (P) as major product on reaction with dil. H_2SO_4

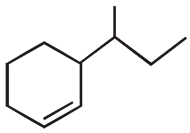


(P)

(A) 

(B) 

(C) 

(D) 

11. Choose the correct option ?

(A) Cyclohexanone forms cyanohydrin in good yield but 2,2,6-trimethyl cyclohexanone does not

(B) Tertiary amines (Me_3N) has higher boiling point than primary amine ($MeNH_2$)

(C) pK_b of aniline is more than that of methyl amine

(D) PCC can oxidise 1° alcohol to aldehyde in anhydrous condition

12. Three isomeric compound **M, N** and **P** ($C_5H_{10}O$) give the following tests :

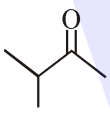
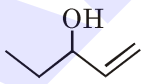
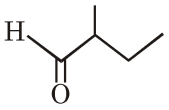
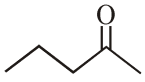
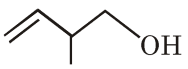
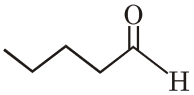
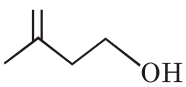
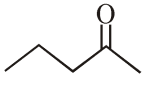
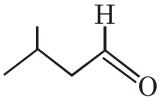
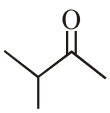
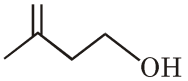
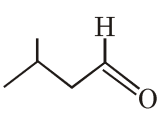
(i) **M** and **P** react with sodium bisulfite to form an adduct

(ii) **N** consumes 1 mol of bromine and also gives trubidity with conc. HCl / anhydrous $ZnCl_2$ after prolong heating

(iii) **M** reacts with excess of iodine in alkaline solution to give yellow crystalline compound with a characteristic smell.

(iv) p-Rosaniline treated with sulphur dioxide developes pink colour on shaking with **P**

The structures of **M, N** and **P,** respectively can be :

	M	**N**	**P**
(A)			
(B)			
(C)			
(D)			

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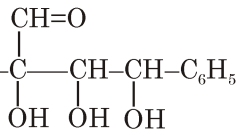
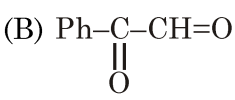
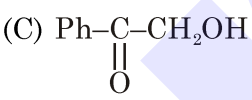
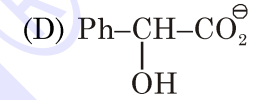
SECTION–I(iii) : Paragraph Type

This section contains **1 paragraph** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

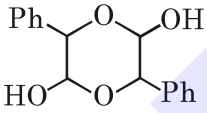
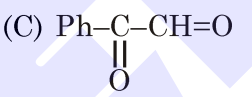
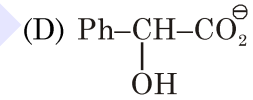
Paragraph for Q.No. 13 to 14

For a long time attempts to prepare compound A ($C_6H_5-\underset{\underset{OH}{|}}{CH}-CH=O$) in basic medium have been failed. Also hydrolysis of its diethyl acetal B ($C_6H_5-\underset{\underset{OH}{|}}{CH}-CH(OCH_2CH_3)_2$) does not give compound A. Actually compound B on acidic hydrolysis gives C :

13. Compound A in basic medium converted into :

(A)  (B)  (C)  (D) 

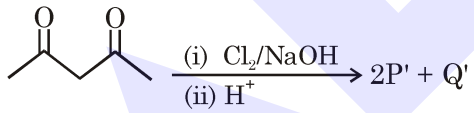
14. Compound C is :

(A) PhCH=CH–CH=O (B)  (C)  (D) 

Matching List Type

This Section contains **2 multiple choice questions. Each question has matching lists.** The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

15. For the following sequence , match the products from list-I with correct common name in list-II.

(i) 

(ii) $P' \xrightarrow[\text{Pyridine}]{CH_3COCl} R'$

(iii) $Ph-CHO + P' \xrightarrow[\text{(ii) } H_3O^+, \Delta]{\text{(i) } R'} S'$

(iv) $\text{Phenol} + Q' \xrightarrow[\text{(ii) } H^+]{\text{(i) KOH}} T'$

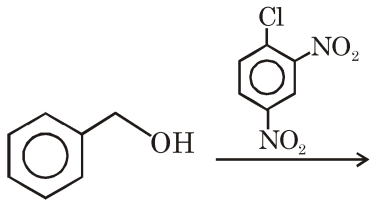
List-I	List-II
(P) P'	(1) Salicyaldehyde
(Q) Q'	(2) Cinnamic acid
(R) T'	(3) Acetic acid
(S) S'	(4) Chloroform

Code :

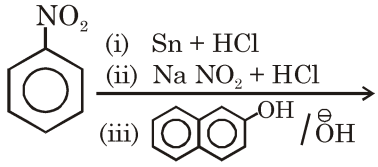
	P	Q	R	S
(A)	4	3	1	2
(B)	3	4	2	1
(C)	4	3	2	1
(D)	3	4	1	2

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16. **List-I** **List-II**

(P)  Product

(1) Product give molish test

(Q)  (T < S°C) Product

(2) Product give violet colour with neutral $FeCl_3$

(R) n-glucose $\xrightarrow[\text{Polymerisation}]{H^+}$ Starch product

(3) L.S. of product give blue colour solution with $FeSO_4$ solution

(S) Amino acid mixture $\xrightarrow{\text{Polymerisation}}$ Polymeric product

(4) Product give Biurate test

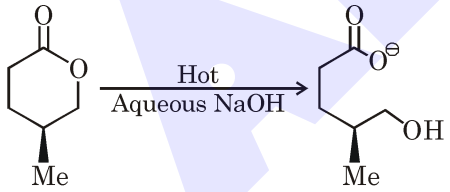
Code :

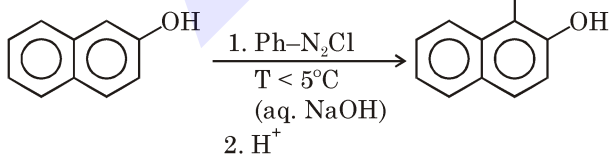
	P	Q	R	S
(A)	3	2,3	1	4
(B)	2	2,3	1	3,4
(C)	3	2,3	1	3,4
(D)	3	2	4	1,3

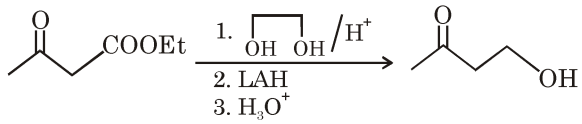
SECTION–III : Integer Type

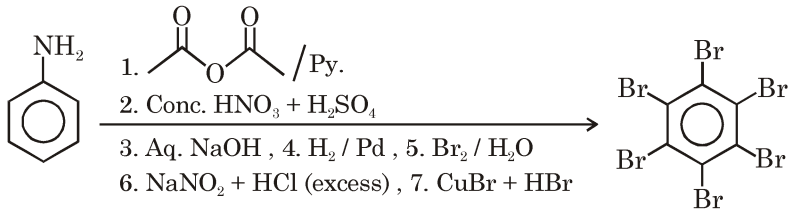
This section contains **4 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

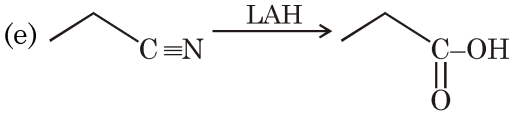
1. Identify reactions correctly matched with their major product ?

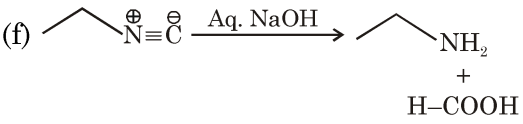
(a) 

(b) 

(c) 

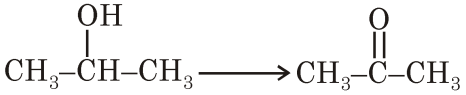
(d) 

(e) 

(f) 

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2. How many of reagents can carry out following transformation :



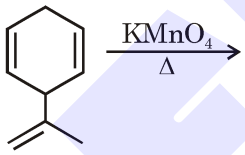
(a) Cu, 300°C (b) O_3 / Zn, H_2O (c) NBS (d) Zn–Hg / HCl

(e) Jones reagent (f) $KMnO_4$, NaOH, Δ (g) $Al(OCMe_3)_3$, 

(h) Collin's reagent (i) (i) TsCl, (ii) DMSO, $NaHCO_3$, Δ (j) $C_5H_5\overset{\oplus}{N}HCrO_3Cl^{\ominus}$ (PCC)

3. Consider all possible carbonyl compounds including stereoisomers of molecular formula $C_5H_{10}O$, All these isomers are independently reacted with NaCN / HCN (**NOTE :** stereoisomers are also reacted separately). The total number of carbonyl compounds that give a racemic product(s) is/are :

4. Total number of moles of CO_2 evolved in following reaction per mole of compound **'M'** is



'M'

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